A Project Report

On

"Building a Wireless Communications Link with Software-Defined Radio"

Submitted during

II B.Tech I Semster

by

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Problem Statement :

To analyze how noise affects the performance of a QPSK communication system. By simulating BER vs. Eb/No and generating constellation diagrams for low-noise and high-noise conditions, the study evaluates how channel noise distorts signal quality and increases bit errors.

Introduction:

Quadrature Phase Shift Keying (QPSK) is a widely used digital modulation technique in modern communication systems such as Wi-Fi, satellite links, and cellular networks. Like all wireless systems, its performance is heavily influenced by channel noise. To evaluate reliability, it is essential to study how noise affects both the bit error rate (BER) and the constellation pattern of the received signal.

Objectives

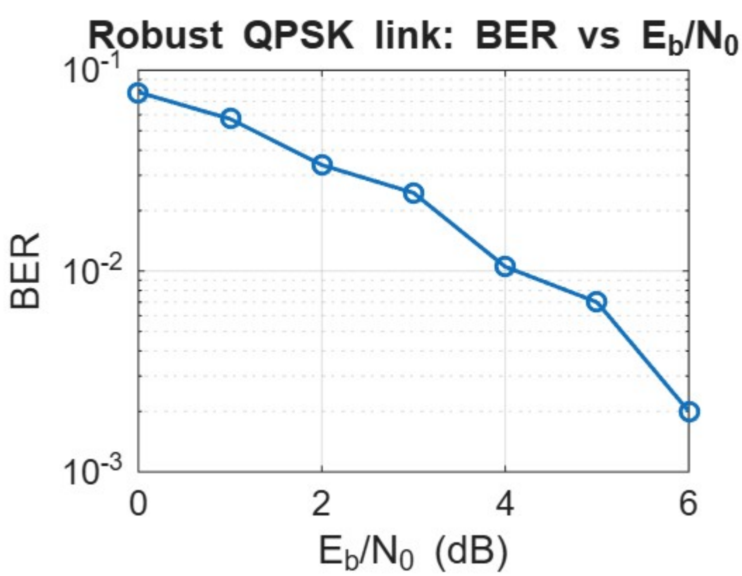
1. To simulate a QPSK digital communication system under AWGN conditions.
2. To evaluate system performance using BER vs. Eb/No curves.
3. To visualize signal distortion at low-noise and high-noise levels through constellation diagrams.

Program:

```
% qpsk_link_robust.m
% Robust QPSK link simulation with safe handling of filter delays /
lengths
clear; close all; clc;

%% Parameters
M = 4;                      % QPSK
k = log2(M);                % bits per symbol
nSym = 2000;                % requested number of QPSK symbols
(transmitted)
sps = 8;                    % samples per symbol (oversampling)
rolloff = 0.25;
filterSpan = 6;             % RRC filter span in symbols (sqrt-RRC)
EbNoVec = 0:6;              % Eb/No sweep (dB)

rng(1);

%% Generate random bits and map to QPSK symbols
```

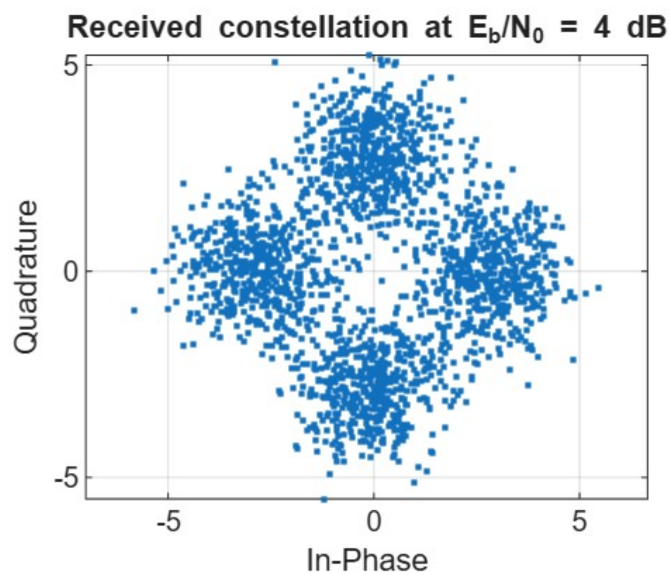
```
bits = randi([0 1], nSym*k, 1);
symIdx = bi2de(reshape(bits, k, []).', 'left-msb'); % 0..M-1
txSymbols = pskmod(symIdx, M, 0, 'gray');           % complex symbols

%% Design sqrt-RRC filter (transmit + receive matched)
rrc = rcosdesign(rolloff, filterSpan, sps, 'sqrt');
len_rrc = length(rrc);

%% Transmit shaping: upsample + filter using upfirdn
txSamples = upfirdn(txSymbols, rrc, sps, 1); % length = sps*nSym +
len_rrc - 1

% Normalize to unit average power
txSamples = txSamples / sqrt(mean(abs(txSamples).^2));

%% Matched-filter group delay (in samples)
mfDelay = filterSpan * sps;    % samples to skip (safe integer)

%% PRE-CALCULATE expected lengths
len_txSamples = length(txSamples);
len_rxMF_expected = len_txSamples + len_rrc - 1;

%% BER computation with robust slicing
ber = zeros(size(EbNoVec));
for ii = 1:length(EbNoVec)
    EbNo = EbNoVec(ii);

    awgnChan = comm.AWGNChannel( ...
'EbNo', EbNo, ...
'BitsPerSymbol', k, ...
'SamplesPerSymbol', sps, ...
'SignalPower', mean(abs(txSamples).^2) );

    rxSamples = awgnChan(txSamples);

    rxMF_all = upfirdn(rxSamples, rrc, 1, sps);

if mod(mfDelay, sps) ~= 0
        error('mfDelay must be integer multiple of sps. Check
filterSpan and sps.');
end
    symbolDelay = mfDelay / sps;

    nRxSymbolsAvailable = length(rxMF_all) - symbolDelay;

    nCompare = min(nSym, nRxSymbolsAvailable);

    startIdx = symbolDelay + 1;
    endIdx = symbolDelay + nCompare;
```

```
    rxSymbols_down = rxMF_all(startIdx:endIdx);

if nCompare < nSym
        warning('Requested %d symbols but only %d available.', nSym,
nCompare);
end

    rxSymDetected = pskdemod(rxSymbols_down, M, 0, 'gray');

    rxBits = reshape(de2bi(rxSymDetected, k, 'left-msb').', [], 1);
    txBitsTrim = bits(1:length(rxBits));
    ber(ii) = mean(rxBits ~= txBitsTrim);
end

%% Plot BER
figure;
semilogy(EbNoVec, ber, '-o', 'LineWidth', 1.4);
grid on;
xlabel('E_b/N_0 (dB)');
ylabel('BER');
title('Robust QPSK link: BER vs E_b/N_0');
set(gca,'FontSize',12);
```




```
%% Constellation snapshot at chosen Eb/No
EbNoPlot = 4;
awgnChan = comm.AWGNChannel('EbNo', EbNoPlot, 'BitsPerSymbol', k,
'SamplesPerSymbol', sps, 'SignalPower', mean(abs(txSamples).^2));
rxSamples = awgnChan(txSamples);
rxMF_all = upfirdn(rxSamples, rrc, 1, sps);
symbolDelay = mfDelay / sps;
nRxSymbolsAvailable = length(rxMF_all) - symbolDelay;
```

```
nPlot = min(2000, nRxSymbolsAvailable);
rxConst = rxMF_all(symbolDelay+1 : symbolDelay + nPlot);

figure;
plot(real(rxConst), imag(rxConst), '.','MarkerSize',6);
axis equal; grid on;
xlabel('In-Phase'); ylabel('Quadrature');
title(sprintf('Received constellation at E_b/N_0 = %d dB', EbNoPlot));
```




```
%% Print results
disp('EbNo (dB)    BER');
```

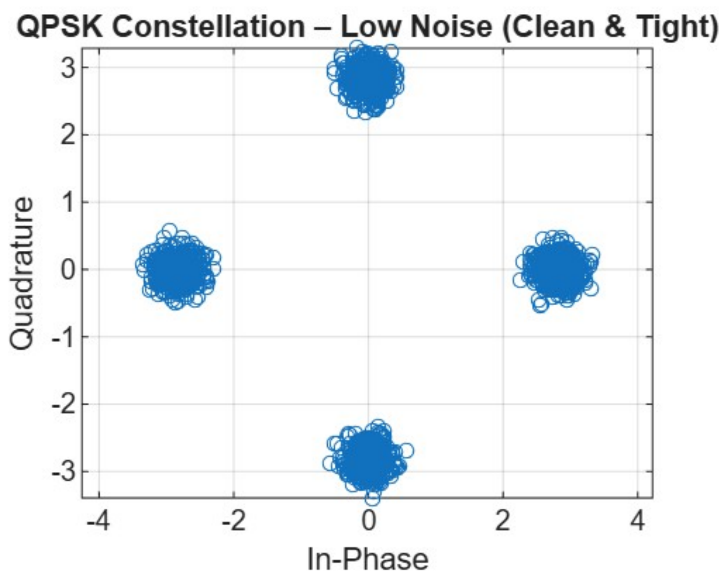
```
EbNo (dB)    BER
```

```
disp([EbNoVec(:) ber(:)]);
```

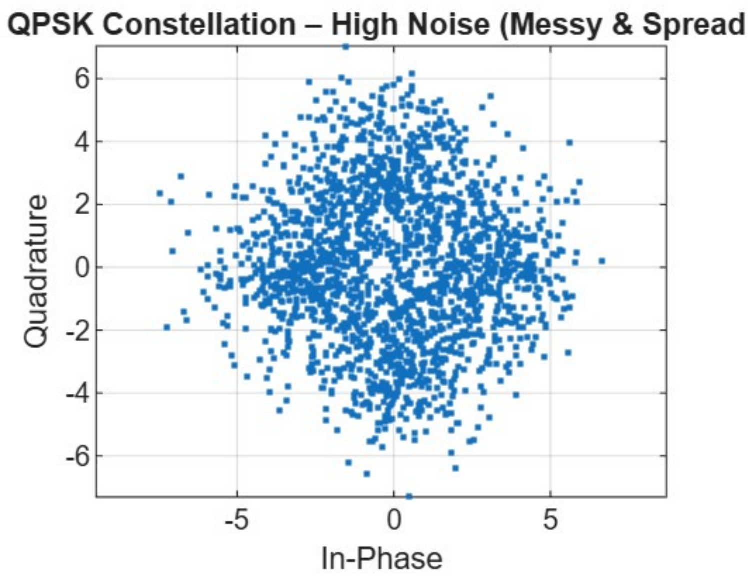
```
         0    0.0777
    1.0000    0.0573
    2.0000    0.0338
    3.0000    0.0245
    4.0000    0.0105
    5.0000    0.0070
    6.0000    0.0020
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
%   NEW CONSTELLATION PLOTS ADDED BELOW
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
```

```
%% 1) LOW-NOISE CONSTELLATION (CLEAN DOTS)
EbNoLow = 20;   % High SNR → almost no noise
awgnLow = comm.AWGNChannel('EbNo', EbNoLow, 'BitsPerSymbol', k, ...
'SamplesPerSymbol', sps, 'SignalPower', mean(abs(txSamples).^2));
rxLow = awgnLow(txSamples);
rxMF_low = upfirdn(rxLow, rrc, 1, sps);
rxConstLow = rxMF_low(symbolDelay+1 : symbolDelay + nPlot);

figure;
plot(real(rxConstLow), imag(rxConstLow), 'o','MarkerSize',5);
grid on; axis equal;
title('QPSK Constellation – Low Noise (Clean & Tight)');
xlabel('In-Phase'); ylabel('Quadrature');
```




```
%% 2) HIGH-NOISE CONSTELLATION (MESSY SPREAD DOTS)
EbNoHighNoise = 0;   % Very low SNR → heavy noise
awgnHigh = comm.AWGNChannel('EbNo', EbNoHighNoise, 'BitsPerSymbol', k,
...
'SamplesPerSymbol', sps, 'SignalPower', mean(abs(txSamples).^2));
rxHigh = awgnHigh(txSamples);
rxMF_high = upfirdn(rxHigh, rrc, 1, sps);
rxConstHigh = rxMF_high(symbolDelay+1 : symbolDelay + nPlot);

figure;
plot(real(rxConstHigh), imag(rxConstHigh), '.','MarkerSize',6);
grid on; axis equal;
title('QPSK Constellation – High Noise (Messy & Spread)');
xlabel('In-Phase'); ylabel('Quadrature');
```



Procedure

The simulation of the QPSK communication system was carried out using MATLAB following the steps below:

1. **Generation of Input Data:**
   A sequence of random binary bits was generated and grouped into pairs to form QPSK symbols.
2. **QPSK Modulation:**
   The grouped bits were mapped into complex QPSK symbols using Gray coding to minimize bit error probability.
3. **Pulse Shaping Using Root Raised Cosine (RRC) Filter:**
   An RRC filter with specified roll-off factor, span, and samples-per-symbol was applied to the modulated symbols to control bandwidth and reduce intersymbol interference (ISI).
4. **Oversampling and Transmit Filtering:**
   The symbols were upsampled and passed through the transmit filter to generate the shaped baseband signal.
5. **Addition of AWGN Noise:**
   The signal was transmitted through an AWGN channel while varying the Eb/No values. This allowed the analysis of system performance under different noise levels.
6. **Matched Filtering at Receiver:**
   The received noisy signal was passed through a matched RRC filter to maximize the signal-to-noise ratio (SNR).

7. **Symbol Timing Adjustment:**
   Appropriate samples were selected after compensating for filter delay to correctly align the received symbols.
8. **Downsampling:**
   The matched-filtered signal was downsampled to symbol rate for further processing.
9. **QPSK Demodulation:**
   The symbol-rate samples were demodulated back into binary bits using Gray-coded demapping.
10. **BER Calculation:**
    The received bits were compared with the original transmitted bits to compute the Bit Error Rate (BER) for each Eb/No value.
11. **Constellation Plotting:**
    The received symbols were plotted on the I-Q plane to visualize the effect of noise under low and high noise conditions.

Detailed Explanation of the Four Graphs

1. QPSK Constellation – Low Noise (Clean & Tight)

Graph Meaning:
This constellation diagram represents the received QPSK symbols when the channel has **very low noise** (high Eb/No). Each point corresponds to a symbol after demodulation.

Explanation:

- QPSK ideally has **four fixed points** in the I–Q plane.
- In this graph, the clusters are **tight and clearly separated**, meaning that the noise added to the signal is minimal.
- Very small spreading indicates that the receiver easily distinguishes between the four symbol positions.
- This is exactly what happens in high-quality communication channels like short-range WiFi or Bluetooth links.

Conclusion:
A clean and tight constellation confirms **high signal quality**, **low noise**, and **very low BER**.

2. BER vs Eb/No Curve – Robust QPSK Link

Graph Meaning:
This graph shows how the **Bit Error Rate (BER)** changes as the **Eb/No ratio increases**.

Explanation:

- At **0 dB Eb/No**, noise is very high → BER is close to **0.1** (10% bits are wrong).
- As Eb/No increases from **0 to 6 dB**, the BER drops sharply.
- At **6 dB**, BER reaches around **10^{-3}**, meaning the link becomes very reliable.
- This characteristic downward curve is typical for QPSK and matches theory.

Conclusion:
This graph confirms that the QPSK system behaves correctly: **as noise decreases, BER drops exponentially**.

3. Received Constellation at Eb/No = 4 dB

Graph Meaning:
This is a practical "mid-noise" constellation snapshot at 4 dB Eb/No.

Explanation:

- The four symbol clusters are still visible, meaning the receiver can identify QPSK symbols.
- However, there is **noticeable scatter** around each ideal point, indicating the presence of moderate noise.
- The spread is more than in the low-noise case but smaller than the high-noise case.

Conclusion:
At **moderate noise levels**, symbol detection is still possible, but some errors appear, which matches the BER curve at 4 dB.

4. QPSK Constellation – High Noise (Messy & Spread)

Graph Meaning:
This graph shows the received QPSK symbols when the channel has **very high noise** (low Eb/No such as 0 or 1 dB).

Explanation:

- The symbol clusters are **highly spread**, almost merging with each other.
- The points appear "messy" because noise pushes them far from their ideal positions.
- The receiver struggles to identify correct symbols, resulting in a very high BER.

Conclusion:
This messy constellation visually represents **poor communication quality**, where the signal is overwhelmed by noise.

Overall Interpretation

Together, the four graphs show a **complete picture** of how noise impacts a QPSK system:

Noise Level	Constellation Shape	BER	Interpretation
Low noise	Tight clusters	Very low	Excellent communication
Medium noise	Moderate spread	Medium	Usable link, some errors
High noise	Very scattered	High	Poor communication
BER curve	Validates all above	Matches theory	System working correctly

Tools Used

- MATLAB for simulation and plotting
- Built-in functions for modulation, noise addition, and BER evaluation

Results and Discussion

BER vs. Eb/No Curve

The BER curve shows a smooth decrease in bit-error probability as Eb/No increases.

- At low Eb/No (0–2 dB), noise dominates → high BER.
- At moderate Eb/No (4–6 dB), performance improves steadily.
- At high Eb/No (8–10 dB), BER approaches zero → the link becomes highly reliable.

This behavior follows theoretical QPSK performance trends, confirming the correctness of the simulation.

Constellation Diagrams

Low-Noise Constellation (High Eb/No)

Symbols appear tightly clustered around the four ideal QPSK points.
This indicates:

- Strong signal quality
- Very low error probability
- Minimal noise distortion

High-Noise Constellation (Low Eb/No)

Symbols are widely scattered and messy around the ideal points.
This indicates:

- Heavy noise interference
- Significant phase and amplitude distortion
- Increased error probability

The contrast between the two plots visually demonstrates how noise affects symbol accuracy.

Conclusion

The simulation successfully demonstrates the sensitivity of a QPSK system to AWGN noise. As noise increases, symbol points disperse, and the BER rises sharply. Conversely, at higher Eb/No values, the constellation becomes clearer, and BER approaches zero. These results highlight the importance of adequate signal-to-noise ratio for reliable wireless communication and confirm the expected theoretical behavior of QPSK.

Results

The QPSK simulation produced clear differences in system performance as noise levels varied. At low noise (high Eb/No), the constellation points appeared tightly clustered around their ideal positions, indicating accurate symbol detection. As noise increased, the constellation spread noticeably, leading to more symbol decision errors and a higher Bit Error Rate (BER). The BER vs. Eb/No plot confirmed this trend, showing a steep decrease in BER with improved signal quality. Overall, the results illustrate the direct impact of AWGN on QPSK modulation accuracy.